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The Effect of Spaced vs. Massed Practice on Word Recall Among University Students

1. Introduction and Background

Every semester, right around midterms and finals, most students end up doing the same thing: cramming. Whether it is an entire chapter the night before an exam or a full unit crammed into a few hours, a lot of us rely on massed, last-minute studying instead of spreading our preparation out over several days. Many students have also noticed that even when cramming gets them through the exam, they forget most of the material within a few days afterward. This is something almost everyone in our group has experienced firsthand, and it made us want to actually test it rather than just assume it based on personal experience.

In our Learning and Memory units this semester, we came across the spacing effect, the well-documented finding that people tend to remember information better when their study time is spread across separate sessions instead of packed into a single block. This felt directly relevant to something every student goes through every exam season, so we decided to design a small experiment to test it ourselves, instead of just relying on the textbook description of the effect.

Beyond satisfying our own curiosity, this topic also has a practical side. If spaced practice really does produce noticeably better recall, that is genuinely useful, actionable information for students juggling multiple courses and back-to-back exams. Past research supports this pattern in different contexts. Cepeda et al. (2006) reviewed hundreds of memory experiments and found a consistent advantage for spaced over massed practice, and Baddeley and Longman (1978) found the same pattern in a completely different, applied setting, where postal workers learning to type retained the skill better when their training sessions were spread out rather than massed together.

We wanted to see whether the same pattern would hold up in a short, real academic memory task with our own peers.

2. Research Question, Hypotheses, and Objective

Research Question: Does distributing study time into separate spaced sessions lead to better word recall than studying the same material in one continuous massed session, among university students?

Hypotheses:

- **H_0 (Null):** There is no significant difference in the number of words recalled between participants who study using massed practice and participants who study using spaced practice.
- **H_A (Alternative):** Participants who study using spaced practice will recall significantly more words than participants who study using massed practice.

Objective: To find out, using a controlled between-subjects experiment, whether spacing out study exposure produces better short-delay recall than massing it into a single session, and to connect the findings back to the spacing effect and to encoding and retrieval theories of memory covered in class.

3. Methodology

This study uses an experimental design rather than a survey or interview, so no questionnaire is attached. The materials used to run the experiment are listed in section 3.3 below.

3.1 Research Design

This is a true experiment, not a correlational study. We are using a between-subjects design, meaning each participant takes part in only one condition, either massed or spaced, not both. The

independent variable is the study schedule (massed vs. spaced), which we are directly manipulating. The dependent variable is the number of words each participant correctly recalls.

3.2 Participants

We plan to recruit around 40–50 university students through convenience sampling, mainly from our own classes, friend groups, and social circles. Participants will be randomly assigned to one of the two conditions, so that each group ends up with roughly 20–25 people. Since the task uses an English word list, participants need to be reasonably comfortable reading and writing in English, we address this as a limitation in our discussion, since it may affect how well the results generalize to students who are less comfortable with English.

3.3 Materials

- A list of 20 unrelated, moderately common English nouns for participants to study
- A short unrelated filler task (simple arithmetic problems), used both between the spaced study blocks and during the delay period before recall
- Blank recall sheets for participants to write down the words they remember
- A stopwatch or phone timer, to keep timing consistent across sessions
- A short informed consent form
- A brief demographic sheet (age, gender, year of study), kept minimal since this is not the focus of the study

3.4 Procedure

Each session is run individually or in a small group setting and takes about 20–25 minutes per participant.

1. Participants read and sign the consent form. They are told this is a "memory study," without being told the specific hypothesis about spacing, so that knowing what we are testing does not change how they behave.
2. Each participant is randomly assigned to either the Massed group or the Spaced group.
3. Massed group: studies the full 20-word list silently, in one continuous 2-minute block.
4. Spaced group: studies the same 20-word list across three separate ~40-second blocks, with about 1 minute of the unrelated filler task between each block. Total study exposure time is equal to the massed group's.

5. Both groups then complete an identical 10–15 minute filler task. This creates a short delay and stops participants from silently rehearsing the list right up until the test.
6. Recall test: participants get 2 minutes to write down as many words from the list as they can remember, in any order.
7. We collect the sheets and score them against a predetermined answer key (minor spelling mistakes are still counted correct if the intended word is clear).

8. Participants are debriefed, told the actual purpose of the study, and thanked for their time.

3.5 Ethical Considerations

Participation is completely voluntary, and participants can withdraw at any point without any consequence. We are not collecting names on the recall sheets, only ID numbers, so individual scores remain anonymous. The only thing not disclosed upfront is the specific hypothesis being tested, which is a standard, minor withholding of information used to prevent participants from changing their natural behaviour, not full deception; everyone receives a complete explanation immediately after the recall test. There is no physical or significant psychological risk involved beyond the normal, mild effort of a short memory task.

4. Psychological Concepts and Operational Definitions

The central concept in this study is the spacing effect: the finding that distributed practice over time produces better retention than the same amount of practice massed into a single block.

- **Massed practice** (operational definition): studying the 20-word list in one continuous, uninterrupted 2-minute block.
- **Spaced practice** (operational definition): studying the same 20-word list across three separate ~40-second blocks, separated by 1-minute unrelated filler activities, with total study time equal to the massed condition.
- **Recall** (operational definition): the number of words, out of 20, a participant correctly writes down during a 2-minute free recall test.
- **Delay** (operational definition): a 10–15 minute unrelated filler task placed between the final study exposure and the recall test, used to prevent active rehearsal and simulate short-term forgetting.